American Computer Science League

2024-2025 ● Contest 1: Short Problems ● Senior Division

1. Computer Number Systems

Convert $9BD_{16}$ to octal.

A. 3535
B. 4575
C. 4675
D. 4775
E. 5775

2. Computer Number Systems

What is the base 10 value of $A5C_{16}$?

A. 2425
B. 2465
C. 2562
D. 2652
E. 2752

3. Recursive Functions

Find $f(10)$ given:

$$f(x) = \begin{cases} 2 & \text{if } x <= 1 \\ f(x-1)+3 & \text{if } x > 1 \text{ and } x \text{ is odd} \\ f(x-1)-1 & \text{if } x > 1 \text{ and } x \text{ is even} \end{cases}$$

A. 8
B. 9
C. 10
D. 11
E. 12

4. Recursive Functions

Find $f(20, 12)$ given:

$$f(x,y) = \begin{cases} f(y-2, x-3)+1 & \text{if } x \geq 12 \\ f(x+2, y-2)-2 & \text{if } 8 \leq x < 12 \\ x+y & \text{if } x < 8 \end{cases}$$

A. 16
B. 17
C. 18
D. 19
E. 20

5. What Does This Program Do?

What is the output of the following program?

```
m = 369
b = int(m / 100)
m = m % 100
q = int(m / 25)
m = m % 25
d = int(m / 10)
m = m % 10
n = int(m / 5)
p = m % 5
output b, q, d, n, p
```

A. 300, 20, 10, 1, 4
B. 30, 20, 1, 1, 4
C. 3, 3, 2, 1, 4
D. 3, 2, 1, 1, 0
E. 3, 2, 1, 1, 4

6. What Does This Program Do?

What is the output of the following program?

```
t = 0
n = 1
while t != 2024
     s = 0
     for y = 1 to n
          s = s + y
     next y
     t = t + s
     n = n + 1
end while
output n-1
```

A. 16
B. 18
C. 20
D. 22
E. 24